

Project Update

Speech-to-Speech LLM Backbone Replacement

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1. Original Plan — Helium Swap on Moshi

The initial goal was to replace the **Helium** language model (7B parameters) inside Kyutai’s Moshi framework with a more accessible open-source LLM. Moshi is a full-duplex speech-to-speech model; Helium drives text generation and reasoning within it.

1.1 Constraint Encountered

Loading the quantized INT8 Moshi checkpoint on Google Colab (T4, 15 GB VRAM) caused CUDA out-of-memory errors because the Mimi audio codec and the LLM together exceed available memory. Colab Pro+ with an A100 (40 GB) is the minimum viable hardware for a direct Helium swap.

Outcome: Direct Moshi modification was not feasible on available hardware. The project pivoted to Mini-Omni.

2. Revised Approach — Mini-Omni + Phi-3.5 Mini Instruct

Mini-Omni (by gpt-omni) is a lightweight open-source speech-to-speech model with a modular LLM backbone, making it a suitable testbed. The default LLM was replaced with **Microsoft Phi-3.5 Mini Instruct** (3.8B parameters).

Item	Detail
Replacement model	<code>microsoft/phi-3.5-mini-instruct</code>
Framework	HuggingFace Transformers 4.41.0 + PEFT
Training hardware	Google Colab (T4, 15 GB VRAM)
Swap script	<code>scripts/lora_finetune.py --backbone ...</code>

3. LoRA Fine-Tuning

LoRA adapters (rank $r = 16$, $\text{lr} = 3 \times 10^{-4}$, batch size 30) were applied to the Phi-3.5 Mini backbone. Three training runs were conducted on a Bengali talkshow audio dataset of 1,000 samples.

3.1 Training Configuration

Parameter	Value
LoRA rank (r)	16
Learning rate	3×10^{-4}
Batch size	30
Trainable parameters	8.9M (of 3,830M total)
Dataset	Bengali talkshow audio, 1,000 samples
Batches per epoch	34

3.2 Training Metrics — Run 1 (3 epochs)

Epoch	Start Loss	End Loss	Accuracy
1	1.8590	1.8308	0.00%
2	1.7472	1.5745	0.00%
3	1.3276	1.0640	0.00%

3.3 Training Metrics — Run 2 (10 epochs)

Epoch	End Loss	Accuracy
1	1.8334	0.00%
2	1.5901	0.00%
3	1.0857	0.00%
4	0.5267	0.00%
5	0.2309	0.00%
6	0.1058	0.00%
7	0.0850	0.00%
8	0.0802	0.00%
9	0.0778	0.00%
10	0.0777	0.00%

3.4 Training Metrics — Run 3 (20 epochs, summary)

Epoch	End Loss	Accuracy
1	1.8329	0.00%
5	0.2088	0.00%
10	0.0775	0.00%
15	0.0709	0.00%
20	0.0514	0.00%

Loss decreased consistently across all runs (from ≈ 1.86 to ≈ 0.05 over 20 epochs), indicating the model was learning the training distribution. However, accuracy remained 0.00% throughout every run.

4. Results Analysis

4.1 Why Accuracy Was 0%

The accuracy metric measured exact token-level match between the model’s predicted next token and the ground truth audio token. This was fundamentally mismatched to the task for two reasons:

- **Wrong prediction target.** LoRA was applied to the text LLM (Phi-3.5 Mini), but the training labels were audio codebook tokens from Mini-Omni’s audio head. The text model’s output vocabulary ($\sim 32,000$ text tokens) has zero overlap with the audio codebook ($\sim 2,048$ entries), so every prediction was marked incorrect regardless of text quality.
- **Modality mismatch.** The dataset contained Bengali talkshow audio features, but no audio-to-text alignment layer was in place. The LoRA adapter reduced cross-entropy loss on the text side, but this did not correspond to producing correct audio tokens.

Accuracy of 0% does not mean the LLM failed to learn — it means the metric was measuring something the LLM was never designed to output directly. The consistently decreasing loss confirms learning was occurring within the text backbone.

4.2 Why No Suitable Dataset Was Found

Proper speech-to-speech fine-tuning requires paired data of the form (*input audio features*, *target audio tokens*), where both sides are encoded by the same codec. No publicly available dataset satisfies this for the following reasons:

- Standard datasets (LibriSpeech, Common Voice) provide raw audio or transcripts, not pre-tokenized audio codebook sequences compatible with Mini-Omni’s codec.
- Bengali speech datasets with sufficient size and quality are extremely scarce. The 1,000-sample talkshow dataset used is far below the tens of thousands of samples needed for audio generation fine-tuning.
- Generating a compatible dataset would require running Mini-Omni’s audio encoder on raw speech to produce paired tokens — a preprocessing pipeline not set up within the project scope.
- Multilingual speech-to-speech datasets that include Bengali are essentially non-existent in the open-source community.

As a result, the fine-tuning adapted the text backbone on an incompatible signal: loss reduced (text-side learning occurred) but audio quality did not improve because no audio-domain supervision was provided.

5. Next Steps — Fixing Mini-Omni Output

- Retrain the audio generation head using Phi-3.5 Mini’s hidden states as input
- Add a vocabulary alignment layer between the Phi-3.5 tokenizer and the audio codebook
- Use a larger speech-text dataset (e.g. LibriSpeech, Common Voice)
- Increase LoRA rank (`r=16` or `r=32`) and target `q_proj/v_proj` layers
- Evaluate using WER on ASR transcript as an objective quality metric

6. Road to Helium Swap on Moshi

6.1 Hardware Requirement

A100 40 GB GPU (Colab Pro+ or Kaggle dual-T4 multi-GPU) is required to load the quantized Moshi checkpoint alongside Mimi.

6.2 Recommended Strategy

1. Load Moshi with Mimi frozen; verify VRAM budget without Helium.
2. Identify Helium’s interface: hidden dimension 4096, output logit space.
3. Insert a learned projection adapter: $3072 \rightarrow 4096$ (Phi-3.5 $\rightarrow$ Moshi internal).
4. Integrate Phi-3.5 Mini by replacing `helium_model` in `moshi/models/loaders.py`.
5. Fine-tune adapter + LoRA layers only; keep Mimi frozen initially.
6. Gradually unfreeze Mimi’s decoder once adapter loss plateaus.

6.3 Alternative LLM Candidates

Model	Params	Notes
Phi-3.5 Mini Instruct	3.8B	Strong reasoning, low VRAM
Llama 3.2 3B	3B	Open, good conversational baseline
Mistral 7B v0.3	7B	Closest to Helium’s size; needs A100
Gemma 2 2B	2B	Most compact; may sacrifice quality

7. Summary

Achieved	Remaining
Phi-3.5 Mini swapped into Mini-Omni	Fix garbled audio (retrain audio head)
LoRA fine-tuning run (10–20 epochs)	Vocabulary/token alignment layer
End-to-end inference pipeline works	Larger speech-text training dataset
Bangla & English prompts tested	Moshi swap on A100 hardware
Moshi VRAM constraints documented	Phi-3.5 → Helium adapter layer